1. Lab 04: DNS Foundations with CML and Wireshark

Lab Name: Lab 04: DNS Foundations with CML and Wireshark **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

The Domain Name System (DNS) is a critical Internet service that translates human-readable domain names (like www.example.com) into numerical IP addresses, allowing computers to locate each other on a network. Sockets provide the foundational programming interface that applications use to communicate over a network via TCP or UDP. Understanding how DNS resolves names and how sockets establish connections is essential for network application development. In this lab, you will explore DNS query and response mechanisms and utilize basic socket programming concepts within your CML environment to bridge the gap between network applications and underlying transport protocols.

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3. Objective / Purpose

To observe DNS resolution traffic and understand the Domain Name System (DNS) by setting up a local DNS server using `dnsmasq`, querying it using `nslookup` (or `dig`), and capturing packets with Wireshark via the Cisco Modeling Labs (CML) UI.

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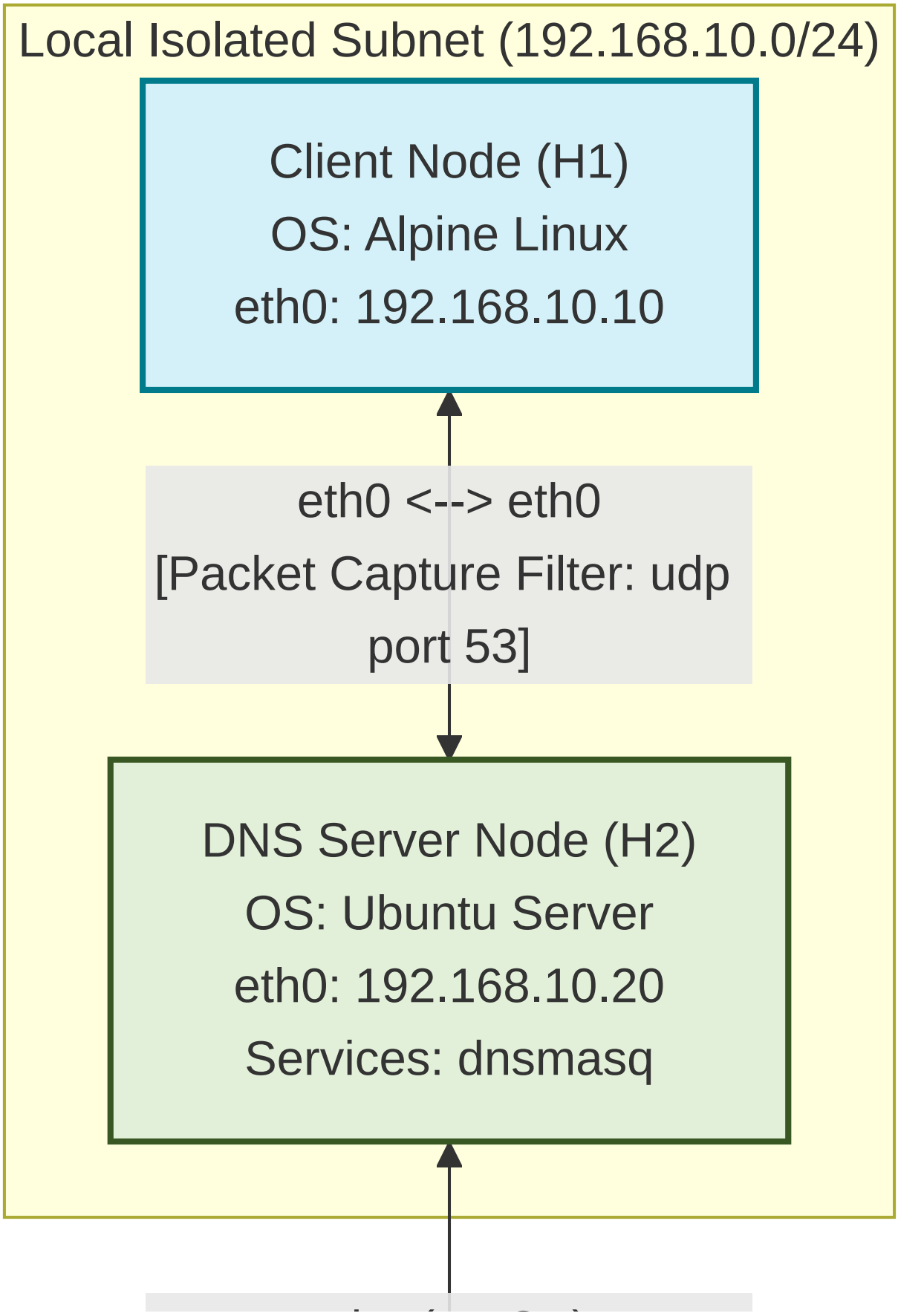
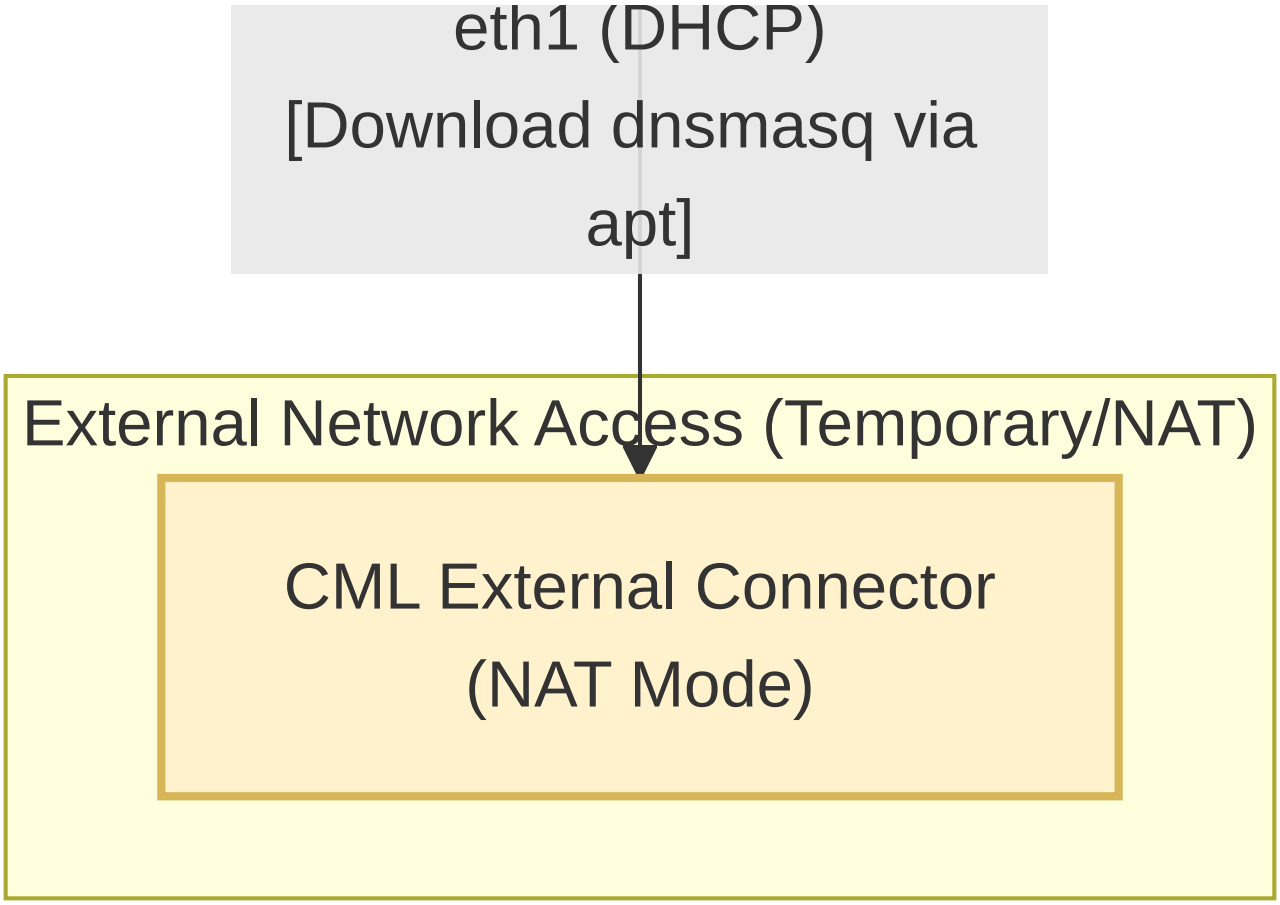
4. Topology Diagram


Local Isolated Subnet (192.168.10.0/24)
Client Node (H1)
OS: Alpine Linux
eth0: 192.168.10.10
eth0 <--> eth0
[Packet Capture Filter: udp port 53]
DNS Server Node (H2)
OS: Ubuntu Server
eth0: 192.168.10.20
Services: dnsmasq



Details:

- Client Node: Alpine Linux (H1)
- DNS Server Node: Ubuntu Server (H2)
- Subnet: 192.168.10.0/24

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 1x Alpine Linux Node (Client), 1x Ubuntu Server Node (DNS Server), 1x CML External Connector (NAT mode)

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6. Configuration / Methodology

IP Addressing Table

Device	Node Type	Interface	IP Address	Subnet Mask
Client (H1)	Alpine Linux	eth0	192.168.10.10	255.255.255.0
Server (H2)	Ubuntu Server	eth0	192.168.10.20	255.255.255.0

Connecting the Server Node to the External World (Internet Access in CML)

Since your nodes reside in a private subnet, they do not have direct internet access to download and install `dnsmasq`. To enable internet access, you must temporarily connect your **Server Node (H2 - Ubuntu)** to CML's **External Connector** in **NAT** mode:

1. Drag an **External Connector** node from the CML palette onto the workspace. Ensure its **Connection Mode** is set to **NAT** in the properties pane.
2. Connect a link from the External Connector to interface `eth1` on the Server Node (leaving `eth0` connected to the Client H1 on your local `192.168.10.0/24` subnet).
3. Start the External Connector and open the Server Node's console.
4. Run the DHCP client or configure the IP parameters manually to enable internet connectivity:

Method A: Automatic Configuration via DHCP (Recommended)

DHCP automatically configures the IP address, default gateway, and DNS servers:

```
sudo dhclient eth1
```

Method B: Manual (Static) Configuration of Gateway and DNS

If assigning static IP parameters on the external bridge subnet (e.g., NAT subnet `192.168.255.0/24` with gateway `192.168.255.1`):

1. **Verify your interface is active**:

   ```
   ip addr show dev eth1
   ```

2. **Configure the Default Gateway**:

   ```
   sudo ip route add default via 192.168.255.1 dev eth1
   ```

3. **Configure the DNS Server**:

   ```
   echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
   ```

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5. Once connected, update the Ubuntu package lists:

   ```
   sudo apt update
   ```

Setting up the DNS Server (`dnsmasq` on Ubuntu Server H2)

First, configure your Server Node to act as a local DNS Server for your environment.

1. Connect to the Server Node console.
2. Install `dnsmasq` on the Ubuntu Server:

```
sudo apt install -y dnsmasq
```

3. Edit the local DNS records. Open `/etc/hosts` and add the following lines at the bottom:

```
192.168.10.20    ns1.example.local
192.168.10.100   www.example.local
```

4. Restart the `dnsmasq` service to apply the changes:

```
sudo systemctl restart dnsmasq
```

Capturing Packets via CML UI

In this lab, you will generate traffic on your Client Node and capture it directly from the link in CML.

1. Open your CML Dashboard.
2. Right-click on the link connecting your Client Node to the network (or directly to the Server Node).
3. Select **Packet Capture**. Set the filter if necessary (e.g., `udp port 53`) and click **Start**.
4. Generate the DNS traffic from your Client Node as instructed in the next sections.
5. Once complete, stop the capture in the CML UI and click **Download PCAP**.
6. Open the downloaded `.pcap` file locally on your computer using Wireshark to answer the questions.

1. nslookup

Let's investigate DNS by examining the `nslookup` command from your Client Node. Point it to your newly created DNS server.

```
nslookup www.example.local 192.168.10.20
```

In its most basic operation, `nslookup` allows the host running `nslookup` to query a specified DNS server for a DNS record. For example, `nslookup` can retrieve a "Type=A" DNS record that maps a hostname to its IP address.

Note: The images below are examples from public queries, but you will perform similar actions against your local `example.local` domain.

```
newworld.cs.umass.edu> nslookup www.nyu.edu
Server:         128.119.240.1
Address:        128.119.240.1#53

Non-authoritative answer:
www.nyu.edu     canonical name = WEB.GSLB.nyu.edu.
Name:   WEB.GSLB.nyu.edu
Address: 216.165.47.12
Name:   WEB.GSLB.nyu.edu
Address: 2607:f600:1002:6113::100
```

Figure 1: the basic nslookup command (example)

We can also use `nslookup` to query for a “TYPE=NS” record, which returns the hostname of an authoritative DNS server.

```
newworld.cs.umass.edu> nslookup -type=NS nyu.edu
Server:         128.119.240.1
Address:        128.119.240.1#53

Non-authoritative answer:
nyu.edu nameserver = ns2.nyu.org.
nyu.edu nameserver = ns4.nyu.edu.
nyu.edu nameserver = ns1.nyu.net.

Authoritative answers can be found from:
ns2.nyu.org     internet address = 128.122.0.76
ns1.nyu.net     internet address = 128.122.0.8
ns4.nyu.edu     internet address = 216.165.87.102
ns4.nyu.edu     has AAAA address 2607:f600:2001:6100::135
```

Figure 2: using nslookup to find authoritative name servers (example)

`nslookup` can also be used to perform a “reverse DNS lookup” by specifying an IP address.

```
kurose@MacBook-Pro-6 ~ % nslookup 128.119.245.12
Server:         75.75.75.75
Address:        75.75.75.75#53

Non-authoritative answer:
12.245.119.128.in-addr.arpa     name = gaia.cs.umass.edu.

Authoritative answers can be found from:
```

Figure 3: using nslookup to perform a “reverse DNS lookup” (example)

Now it’s time for you to test drive it yourself. Start your packet capture in CML, then run the following on your Client Node:

1. Run `nslookup` to obtain the IP address of `www.example.local` using your server `192.168.10.20`. What is the IP address returned?
2. What is the IP address of the DNS server that provided the answer to your command in question 1 above?
3. Use the `nslookup` command to query the `example.local` domain. *Note: `dnsmasq` might not return a full NS record by default without extra configuration, but observe the response.* What information is returned?

2. The DNS cache on your Client Node

Most desktop operating systems keep a cache of recently retrieved DNS records. However, because your Client Node H1 is running **Alpine Linux**, it does not utilize a local DNS caching daemon (such as `systemd-resolved` or `nscd`) by default.

On Alpine Linux, every DNS query is sent directly to the configured DNS server without checking a local cache. Therefore, **no cache flushing commands are needed** on the Client Node for this lab.

3. Tracing DNS with Wireshark

Now let’s look at the captured DNS messages.

- (Skip if using Alpine Linux Client) Ensure your DNS cache is cleared on the Client Node.
- Start packet capture in the CML UI on the link.
- Perform a lookup for `www.example.local` and another external lookup if your lab has internet access (e.g., `nslookup www.cisco.com 192.168.10.20`).
- Stop packet capture and download the PCAP. Open it in Wireshark.

Answer the following questions based on your capture:

5. Locate the first DNS query message for `www.example.local`. What is the packet number in the trace for the DNS query message? Is this query message sent over UDP or TCP?
6. Now locate the corresponding DNS response. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?
7. What is the destination port for the DNS query message? What is the source port of the DNS response message?
8. To what IP address is the DNS query message sent?
9. Examine the DNS query message. How many “questions” does this DNS message contain? How many “answers” does it contain?
10. Examine the DNS response message to the initial query message. How many “questions” does this DNS message contain? How many “answers” does it contain?

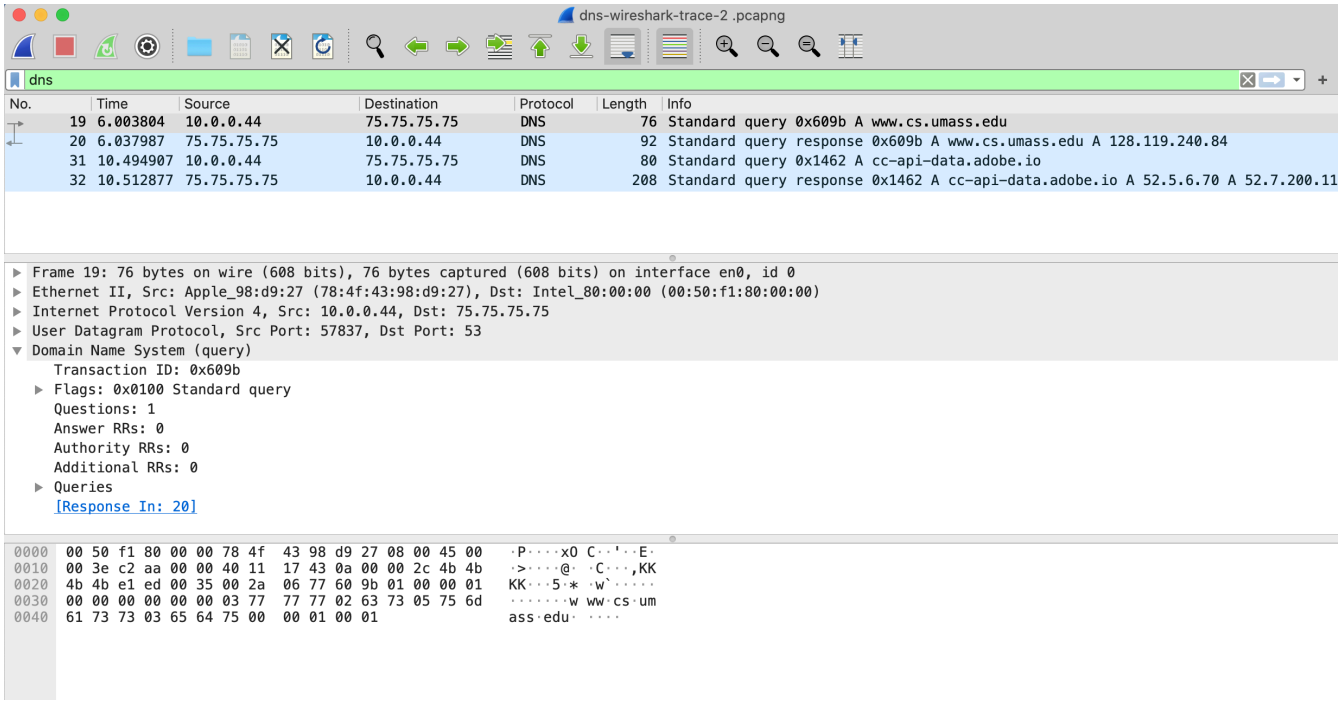

Figure 5: Wireshark trace for nslookup (example)

Examine a DNS query that your server had to forward or resolve recursively (if internet is available), or simply examine your local queries:

12. What is the destination port for the DNS query message? What is the source port of the DNS response message?
13. To what IP address is the DNS query message sent? Is this the IP address of your configured local DNS server (`192.168.10.20`)?

14. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?
15. Examine the DNS response message to the query message. How many “questions” does this DNS response message contain? How many “answers”?

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7. Verification & Testing

1. Screenshot of the CML UI showing the packet capture running.
2. Screenshot of Wireshark showing DNS Standard Query and Response from your downloaded `.pcap`.
3. Provide answers to the questions listed in the text above.

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Verified that DNS uses UDP port 53 and maps hostnames to IP addresses. Successfully set up a local `dnsmasq` server, generated DNS traffic, captured packets via CML, and analyzed the downloaded `.pcap` using Wireshark.

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10. What to Hand In

- The Lab YAML file with your name, date, name of the lab at the top of the file in comments.
- A PDF with annotated screenshots showing the lab running, as well as screenshots of your Wireshark analysis, and the answers to the questions.